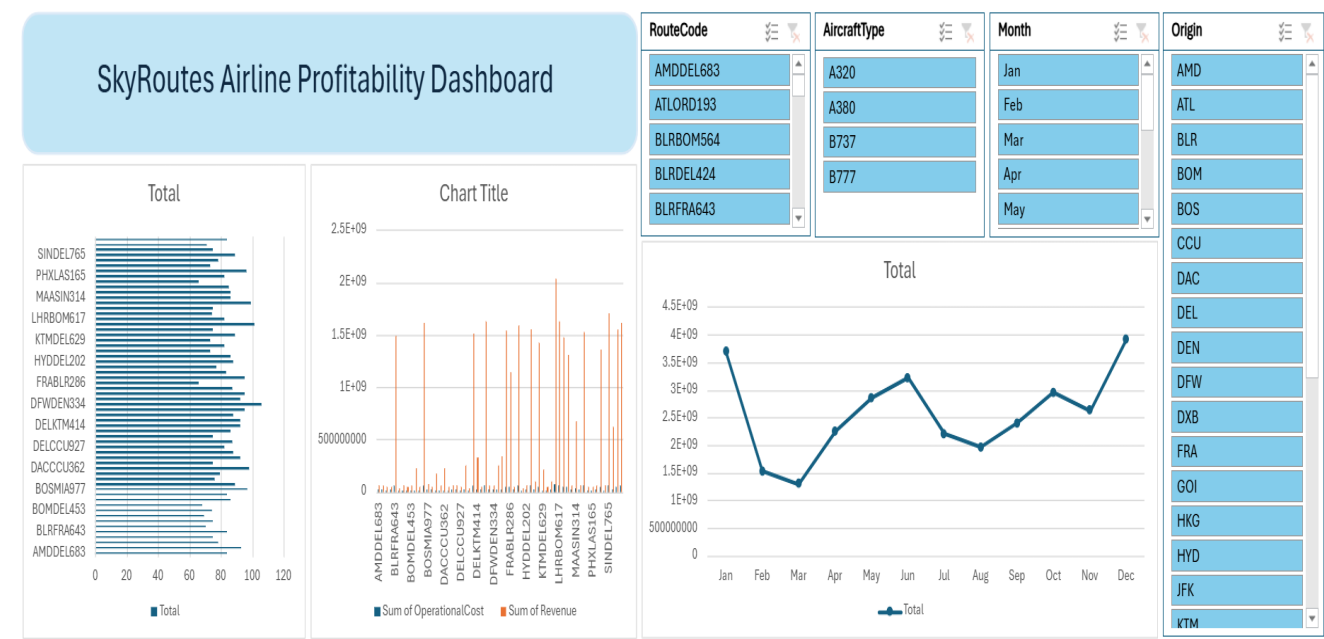


SkyRoutes Airline Profitability Dashboard

A Business Intelligence Analyst Case Study

This case study presents a business-oriented airline analytics project focused on identifying profitable airline routes using SQL and Excel dashboarding techniques. The analysis was performed on a dataset containing 5,000 flight records and aimed to uncover operational insights, revenue trends, occupancy patterns, and profitability metrics.



1. Project Overview

The objective of this project was to evaluate airline route performance and identify high-performing as well as underperforming routes through structured data analysis. SQL queries were used to process and analyze flight data, while Microsoft Excel was used to build an interactive dashboard for visualization and reporting purposes.

2. Business Objectives

- Identify the top 10 most frequent airline routes.
- Calculate average revenue, operational cost, and profit per route.
- Detect routes generating negative profitability.
- Analyze seat occupancy percentage for operational efficiency.
- Study monthly profit trends and seasonal patterns.
- Compare domestic and international route profitability.
- Rank routes based on revenue generated per minute.

3. Dataset Description

The dataset consisted of 5,000 airline flight records containing information such as flight identifiers, route codes, departure and destination airports, flight duration, aircraft type, seat availability, seats sold, revenue generated, and operational costs. This structured dataset enabled comprehensive financial and operational analysis.

4. SQL Analysis & Findings

The SQL analysis revealed that several airline routes operated at significantly higher frequencies, indicating strong passenger demand and business importance. Revenue and operational cost analysis showed that international routes generally generated higher profitability due to premium pricing and longer travel distances. The occupancy analysis demonstrated that routes with higher seat occupancy rates consistently achieved stronger profit margins. A few underperforming routes experienced negative profitability due to elevated operational expenses and comparatively lower passenger demand. Monthly trend analysis identified fluctuations in profitability throughout the year, suggesting the impact of seasonal travel patterns on airline revenue performance. The revenue-per-minute calculation further highlighted operationally efficient routes capable of maximizing returns within shorter flight durations.

5. Dashboard Features

The Excel dashboard was designed to provide an interactive and visually intuitive representation of airline performance metrics. The dashboard included KPI cards, line charts, bar charts, revenue versus cost comparisons, occupancy analysis, and slicer-based filtering for route codes, aircraft types, months, and origin airports. The dashboard enabled dynamic business analysis and improved decision-making capabilities through real-time visual interpretation of operational data.

6. Skills Demonstrated

This project demonstrated practical Business Intelligence Analyst skills including SQL querying, data cleaning, data aggregation, dashboard development, KPI reporting, trend analysis, data visualization, and business insight generation. It also reflected the ability to transform raw data into actionable business recommendations.

7. Conclusion

The SkyRoutes Airline Profitability Dashboard successfully illustrated how data analytics and business intelligence tools can support strategic decision-making in the aviation industry. Through structured SQL analysis and dashboard visualization, the project identified profitable routes, operational inefficiencies, and performance trends that can help airlines optimize route planning, improve profitability, and enhance operational efficiency.